

21-Day Backend Developer Roadmap (Node.js, Express.js, MongoDB)

WEEK 1 — Node.js Core + Express.js Basics (Days 1–7)

Day 1 — JavaScript for Backend

- let, const, var
- Functions, callbacks
- Promises, async/await
- ES modules

Tutorial: JavaScript Crash Course — Dave Gray

Resource: MDN Docs

Day 2 — Introduction to Node.js

- What is Node
- Event Loop
- npm basics
- CommonJS vs ES Modules

Tutorial: Node.js Crash Course — Traversy Media

Resource: Node.js Docs

Day 3 — Node.js Core Modules

- fs module
- path module
- http module
- events
- streams

Tutorial: Net Ninja Node.js Playlist

Day 4 — Express.js Basics

- Routing
- Middleware
- req & res object
- Status codes

Tutorial: Express Crash Course — Traversy Media

Docs: Express Official Docs

Day 5 — Express.js Advanced

- Custom middleware
- Router
- MVC structure
- Error handling

Tutorial: Express Tutorial — Net Ninja

Day 6 — Build REST API

- CRUD operations
- Postman basics

Tutorial: REST API Crash Course — Traversy Media

Day 7 — Mini Project

- Notes API using Express

WEEK 2 — MongoDB + Mongoose + Authentication (Days 8–14)

Day 8 — MongoDB Basics

- Documents, Collections
- CRUD operations
- Filters & projections

Tutorial: MongoDB Tutorial — Mosh

Day 9 — MongoDB Atlas + Compass

- Create cluster
- Connect to Node

Tutorial: MongoDB Atlas Guide

Day 10 — Mongoose Basics

- Schemas
- Models
- Validation

Tutorial: Mongoose Crash Course — Traversy Media

Day 11 — MongoDB Advanced

- Population
- Aggregation
- Indexes

Tutorial: Mongoose Relationships Guide

Day 12 — JWT Authentication

- bcrypt hashing
- JWT token generate/verify

- Login routes

Tutorial: JWT Auth Tutorial — Traversy Media

Day 13 — Authorization

- Protected routes
- Role-based access

Day 14 — Mini Project

- Auth + Notes API

WEEK 3 — Production + Real Project + Deployment (Days 15–21)

Day 15 — Security

- CORS
- Helmet
- Rate limiter
- XSS protection

Tutorial: Node Security Tutorial

Day 16 — API Testing

- Postman collection
- Environment variables
- Writing test cases

Tutorial: Postman Beginner Guide

Day 17 — Deployment

- Render/Railway

- Mongo Atlas connection

Tutorial: Node Deployment — Traversy Media

Day 18 — Logging

- Morgan
- Winston

Day 19 — Final Project

- Complete backend with CRUD + Auth

Day 20 — Advanced Features

- Pagination
- File upload (multer)
- Cloudinary storage

Day 21 — Documentation

- README
- API endpoints
- Swagger basics